ELECTRONICS RETAIL SALES ANALYSIS DASHBOARD

Using Microsoft Power BI – Project Report

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Tool Used: Microsoft Power BI Desktop

ELECTRONICS RETAIL SALES ANALYSIS DASHBOARD

1. Introduction

The Electronics Retail Sales Analysis Dashboard is developed using Microsoft Power BI to analyse the sales performance of an electronics retail business. It provides insights into revenue, profit, order volume, product categories, store performance, and geographical sales distribution.

The dashboard uses a star-schema data model consisting of one fact table and eight dimension tables. DAX measures and interactive visualisations are used to convert raw sales data into meaningful business insights. This helps management understand sales trends, identify profitable product categories, evaluate store performance, and make informed business decisions.

2. Objectives

The main objectives of the Electronics Retail Sales Analysis Dashboard are:

1. To analyse the overall sales performance through revenue, profit, quantity, and order-related metrics.
2. To identify the product categories contributing the highest revenue and profit.
3. To compare the sales performance of different retail stores.
4. To analyse monthly revenue trends and understand changes in sales over time.
5. To evaluate revenue distribution across different store types.
6. To identify cities and states contributing to overall sales.
7. To provide interactive filters for analysing sales based on product category, year, customer segment, store type, and state.

3. Dataset Description

The dataset consists of nine CSV files containing information about sales transactions, products, customers, stores, employees, and geographical locations.

Table Name	Description
Fact Sales	Contains transaction-level sales information, including quantity, unit price, total amount, cost amount, and profit amount.
Dim Product	Contains product details such as product ID and product category.
Dim Customer	Contains customer information and customer segments.
Dim Store	Contains store details and store type information.
Dim Salesperson	Contains salesperson-related information.

Table Name	Description
Dim Date	Contains date-related information used for time-based analysis.
Dim Order	Contains order-related details.
Dim City	Contains city information and its corresponding state ID.
Dim State	Contains state-related geographical information.

The Fact Sales table acts as the central table, while the remaining eight dimension tables provide additional information for analysing sales from different perspectives.

4. Data Preparation

Data preparation was performed using Power Query Editor in Power BI to ensure data consistency and accuracy.

The following steps were carried out:

1. **Data Import:** Imported all nine CSV files into Power BI and renamed the project as *Electronics Retail Sales.pbix*.
2. **Data Type Conversion:** Changed the data types of columns according to their requirements. Numerical fields were converted into appropriate numeric formats, while date fields were assigned the Date data type.
3. **Boolean Conversion:** Converted text values represented by 't' into Boolean values wherever applicable.
4. **Data Cleaning:** Checked the tables for missing values, incorrect data types, and inconsistencies.
5. **Preserving Blank Values:** Blank salesperson IDs were retained instead of replacing them with artificial values.
6. **Data Validation:** Verified the imported tables and checked whether the data was suitable for creating relationships and DAX measures.

These preparation steps helped maintain data consistency and supported accurate sales analysis.

5. Data Modelling and Relationships

A star-schema data model was created by connecting the central Fact Sales table with the dimension tables.

The following relationships were established:

Fact Table / Source	Dimension Table / Target	Relationship
Fact Sales[product_id]	Dim Product[product_id]	Many-to-One
Fact Sales[customer_id]	Dim Customer[customer_id]	Many-to-One
Fact Sales[store_id]	Dim Store[store_id]	Many-to-One
Fact Sales[salesperson_id]	Dim Salesperson[salesperson_id]	Many-to-One
Fact Sales[date_id]	Dim Date[date_id]	Many-to-One
Fact Sales[order_id]	Dim Order[order_id]	Many-to-One
Dim City[state_id]	Dim State[state_id]	Many-to-One
Dim Store[city_id]	Dim City[city_id]	Many-to-One

All relationships were configured as active relationships with single-direction cross-filtering.

This data model allows users to analyse sales performance across different dimensions, including products, customers, stores, salespersons, dates, and geographical locations.

6. DAX Measures

DAX measures were created to calculate important sales performance indicators.

1. Total Quantity

Total Quantity = SUM('Fact Sales'[quantity])

Calculates the total quantity of products sold across all transactions.

2. Total Revenue

Total Revenue = SUM('Fact Sales'[total_amount])

Calculates the overall revenue generated from sales.

3. Total Cost

Total Cost = SUM('Fact Sales'[cost_amount])

Calculates the total cost associated with the products sold.

4. Total Profit

Total Profit = SUM('Fact Sales'[profit_amount])

Calculates the total profit generated from sales transactions.

5. Average Unit Price

Average Unit Price = AVERAGE('Fact Sales'[unit_price])

Calculates the average selling price per unit across the sales records.

6. Total Transactions

Total Transactions = COUNT('Fact Sales'[sales_id])

Counts the total number of sales records available in the Fact Sales table.

7. Total Orders

Total Orders = DISTINCTCOUNT('Fact Sales'[order_id])

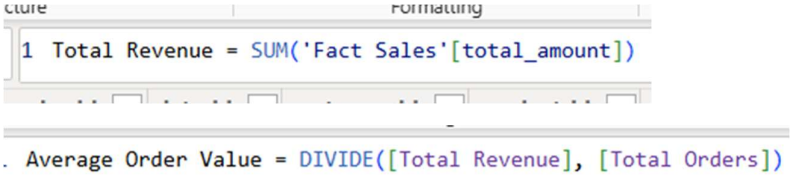
Calculates the total number of unique orders placed by customers.

8. Average Order Value

Average Order Value = DIVIDE([Total Revenue], [Total Orders])

Calculates the average revenue generated per order by dividing total revenue by the total number of unique orders.

Formatting: Monetary measures were formatted using Indian Rupee (₹) currency formatting, while quantity, transactions, and orders were formatted as whole numbers.



7. Data Components and Visuals Used

The dashboard contains different visualisations to analyse the retail sales performance.

1. KPI Cards

Four KPI cards were created to display the major performance indicators:

- Total Revenue
- Total Profit
- Total Orders
- Total Quantity

These cards provide a quick overview of the overall business performance.

2. Clustered Column Chart: Revenue by Product Category

- **X-axis:** Product Category
- **Y-axis:** Total Revenue

This chart compares revenue generated by different product categories and helps identify the categories contributing the highest sales.

3. Treemap: Profit by Product Category

- **Group:** Product Category
- **Values:** Total Profit

The treemap represents the contribution of each product category to overall profit. Larger sections indicate higher profit contributions.

4. Clustered Bar Chart: Revenue by Store

- **Y-axis:** Store
- **X-axis:** Total Revenue

This chart compares revenue generated by individual stores and helps identify high-performing and low-performing locations.

5. Line Chart: Monthly Revenue Trend

- **X-axis:** Year and Month Name
- **Y-axis:** Total Revenue

This chart displays monthly revenue trends and helps identify changes in sales performance over time.

6. Doughnut Chart: Revenue Distribution by Store Type

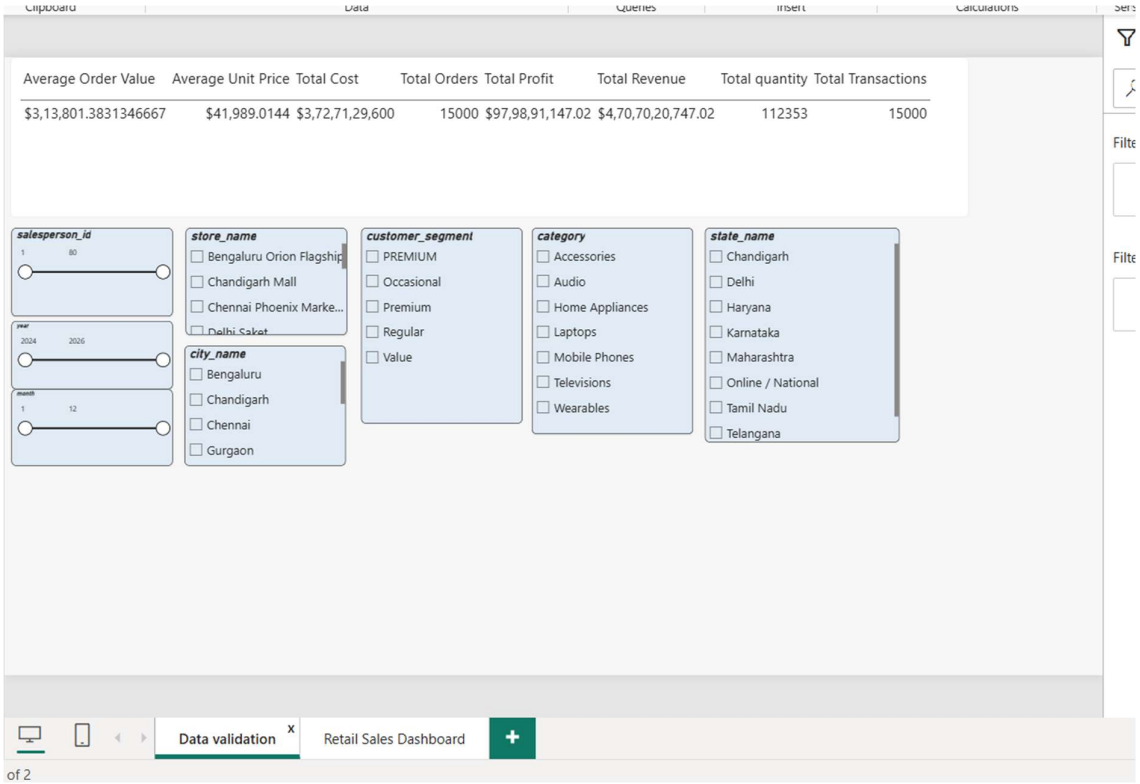
- **Legend:** Store Type
- **Values:** Total Revenue

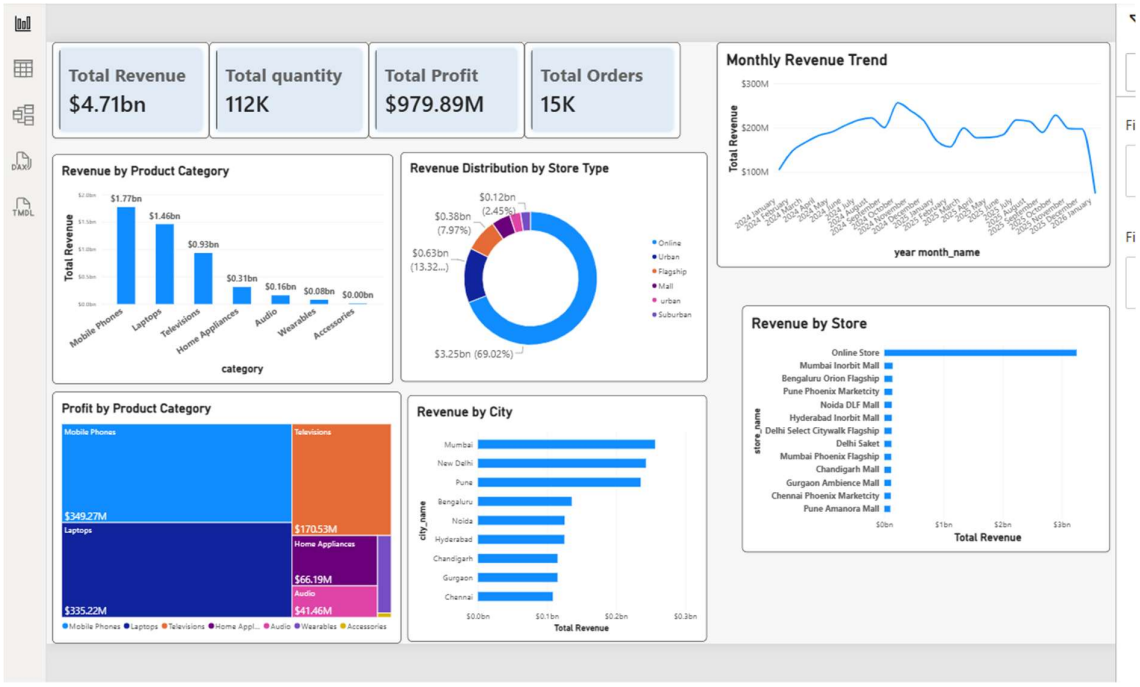
This visual represents the percentage contribution of different store types to total revenue.

7. Clustered Bar Chart: Revenue by City

- **Y-axis:** City
- **X-axis:** Total Revenue

This chart compares revenue across different cities. The category "Online" was excluded from this visual to focus on geographical store-level performance.



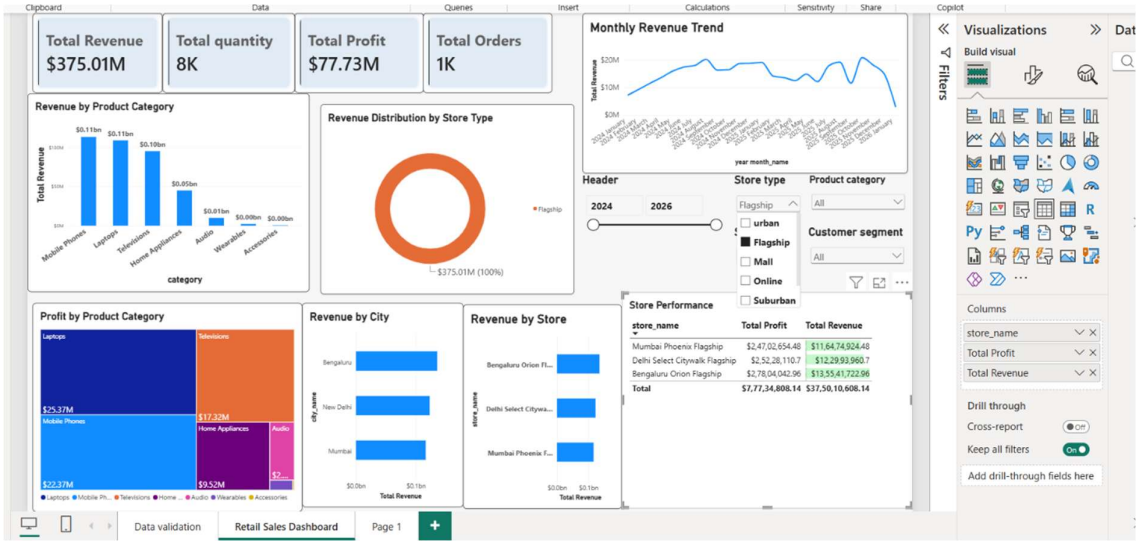


8. Slicers

The following slicers were added to make the dashboard interactive:

- Product Category
- Year
- Customer Segment
- Store Type
- State

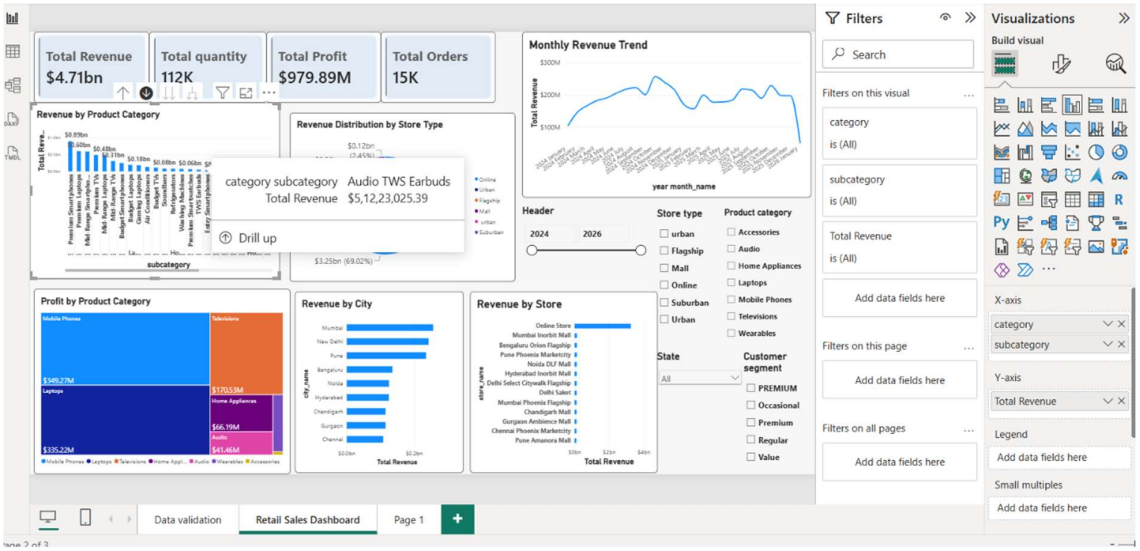
These slicers allow users to filter the dashboard and analyse specific segments of the retail business.

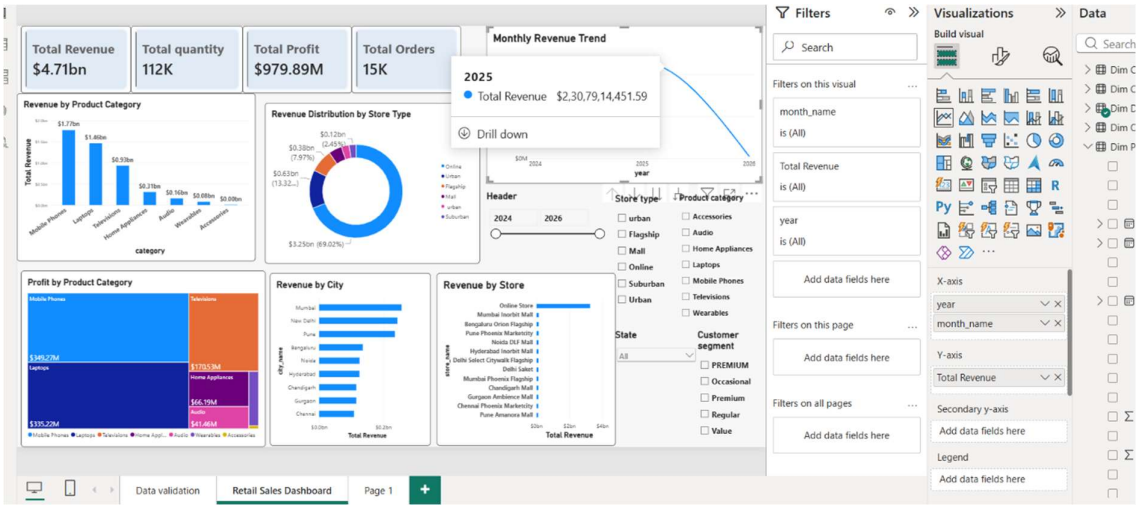


9. Drill-Down Analysis

Drill-down functionality was implemented to provide detailed analysis:

- **Product Category → Subcategory:** Allows users to explore product-level sales performance.
- **Year → Month:** Allows users to examine revenue trends at a monthly level.



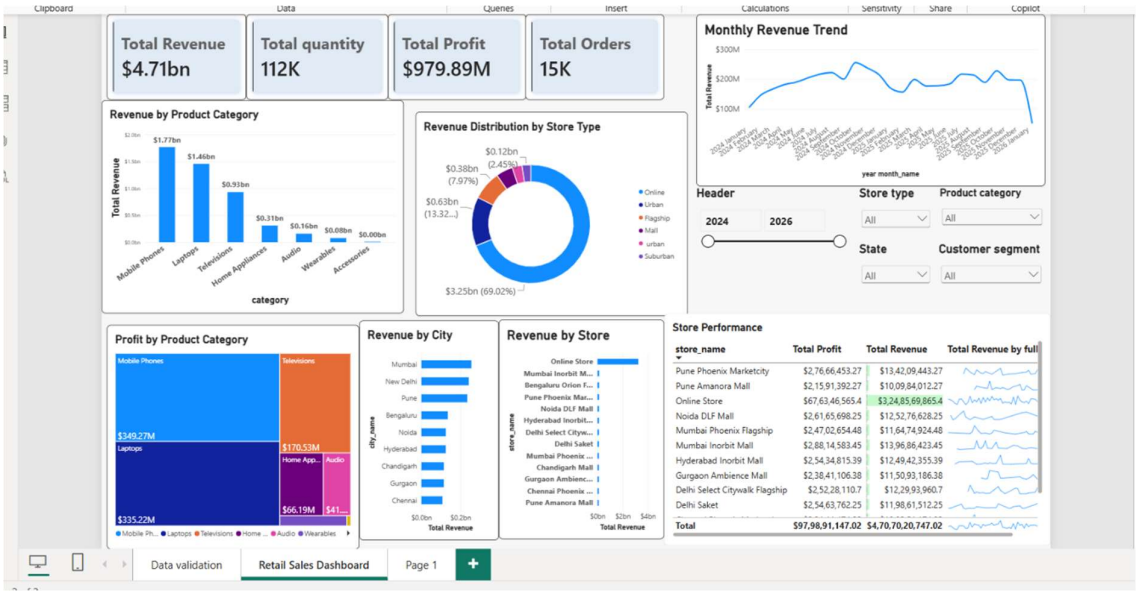


10. Store Performance Table

A table was created to display:

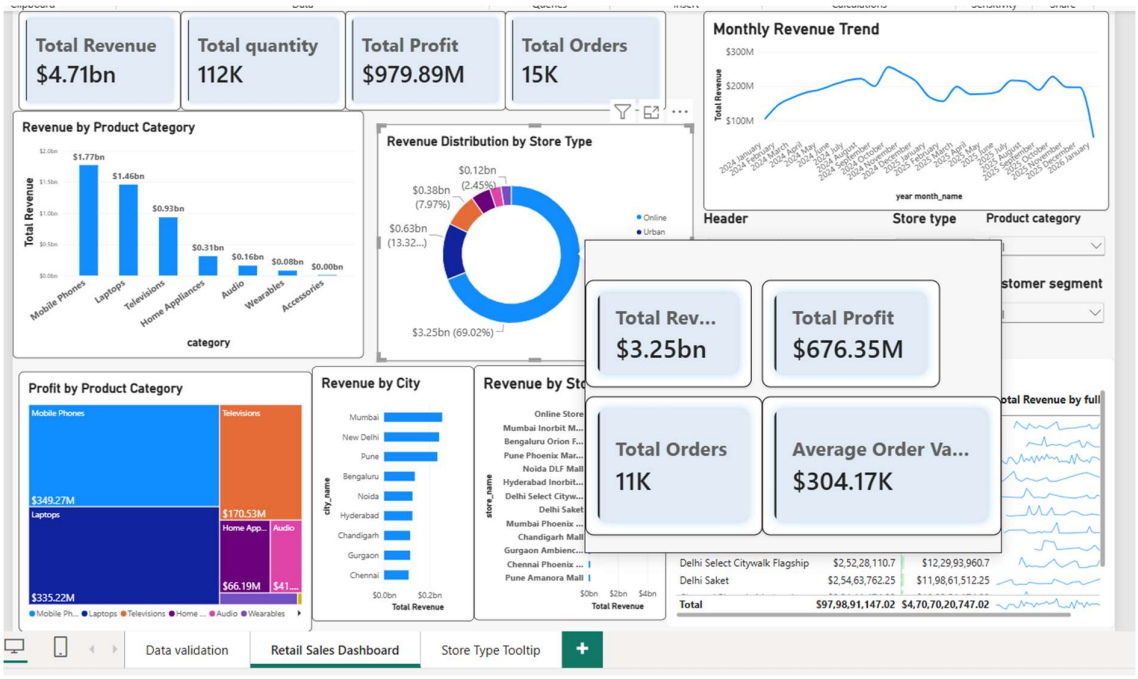
- Store Name
- Total Revenue
- Total Profit

Revenue data bars and sparklines were added to improve the presentation and comparison of store performance.



11. Store Type Tooltip Page

A separate tooltip report page was created with four KPI cards to display additional information when users hover over the store-type-related visual.



12. Visual Interactions

Interactions between charts, slicers, and other dashboard components were configured to support dynamic filtering and cross-analysis.

8. Key Findings

The dashboard provides several opportunities to evaluate the performance of the electronics retail business.

1. **Revenue Performance:** The Total Revenue KPI provides an overview of the overall sales generated by the business.
2. **Profitability Analysis:** The Total Profit KPI and profit-by-category treemap help identify the contribution of different product categories towards business profitability.
3. **Product Category Performance:** The revenue-by-category chart helps compare product categories and identify the major contributors to sales.
4. **Store Performance:** The revenue-by-store chart and store performance table help compare individual stores based on revenue and profit.
5. **Monthly Sales Trends:** The monthly revenue line chart helps identify fluctuations in sales and understand the business's performance over time.
6. **Store Type Analysis:** The doughnut chart helps compare the revenue contribution of different store types.

7. **Geographical Performance:** The city-wise revenue chart provides insights into sales distribution across different geographical locations.
8. **Customer Segment Analysis:** The customer segment slicer allows management to examine sales performance across different customer groups.
9. **Order Analysis:** Comparing Total Transactions and Total Orders helps understand the difference between individual sales records and unique customer orders. A single order may contain multiple sales records.

Note: Actual numerical findings, such as the highest-revenue category, most profitable store, and highest-performing month, should be added after reviewing the completed Power BI dashboard. These values should be taken directly from the dashboard rather than assumed.

9. Conclusion

The Electronics Retail Sales Analysis Dashboard provides a structured approach to analysing retail business performance using Microsoft Power BI. By integrating nine datasets, establishing relationships, creating DAX measures, and developing interactive visualisations, the dashboard converts raw sales information into meaningful insights.

It helps management evaluate revenue, profitability, product category performance, store efficiency, and geographical sales distribution. Overall, the dashboard supports data-driven decision-making and helps identify areas that require attention to improve retail business performance.